

Norm and Trace.

Def. Let K be a finite extension of F , and let $\alpha \in K$.

Let L be a Galois extension of F containing K , and let $|\cdot| \leq \text{Gal}(L/F)$ corresponding to K .

Define $N_{K/F}: K \rightarrow F; \alpha \mapsto \prod_{\sigma \in S} \sigma(\alpha)$ where $S = \{\tau: K \rightarrow F \mid \tau \text{ is embedding}\}$

and $\text{Tr}_{K/F}: K \rightarrow F; \alpha \mapsto \sum_{\sigma \in S} \sigma(\alpha)$ //

In particular, if K is Normal Extension over F , then $S = \text{Gal}(K/F)$.

Prop. To simplify, we assume that K/F is Normal. Then,

1). The Norm and Trace are well-defined.

2). The Norm preserves multiplication, and
The Trace preserves addition.

3). For $\alpha \in K$, suppose that $m_\alpha(x) = x^d + a_{d-1}x^{d-1} + \dots + a_1x + a_0 \in F[x]$ is the minimal of α .

Then, d divides $[K:F] = n$.

$$N_{K/F}(\alpha) = (-1)^n \cdot a_0^{\frac{n}{d}} \quad \text{and} \quad \text{Tr}_{K/F}(\alpha) = -\frac{n}{d} \cdot a_{d-1}.$$

Proof. Given $\tau \in G(K/F)$, and $\alpha \in K$, since $\tau \circ \sigma$ are Automorphism

$$\tau(N_{K/F}(\alpha)) = \tau\left(\prod_{\sigma \in \text{Gal}(K/F)} \sigma(\alpha)\right) = \prod_{\sigma \in \text{Gal}(K/F)} (\tau \circ \sigma)(\alpha) = \prod_{\sigma \in \text{Gal}(K/F)} \sigma(\alpha) = N_{K/F}(\alpha),$$

So the norm is fixed by $G(K/F)$, it means the norm is contained in F , Similarly trace.

And, given $\alpha, \beta \in K$,

$$N_{K/F}(\alpha\beta) = \prod_{\sigma \in \text{Gal}(K/F)} \sigma(\alpha\beta) = \left(\prod_{\sigma \in \text{Gal}(K/F)} \sigma(\alpha)\right) \cdot \left(\prod_{\sigma \in \text{Gal}(K/F)} \sigma(\beta)\right) = N_{K/F}(\alpha) \cdot N_{K/F}(\beta), \quad \text{similarly, trace.}$$

Now, let $m_\alpha(x)$ be the minimal polynomial of α . Then,

$[K:F(\alpha)] \cdot [F(\alpha):F] = [K:F] = n$, and $[F(\alpha):F] = \deg m_\alpha = d$, so $d \mid n$.

So, $|\text{Gal}(K/F)| = n$, and $\sigma \in \text{Gal}(K/F)$ conjugates α to root of m_α , therefore there are n/d repetition in the product. Consequently,

$$N_{K/F}(\alpha) = (-1)^n \cdot a_0^{\frac{n}{d}}.$$

And,

$$\text{Tr}_{K/F}(\alpha) = -\frac{n}{d} \cdot a_{d-1}$$

(observe $m_\alpha(x) = (x - \alpha_1) \cdots (x - \alpha_d)$).

Example. For $K = \mathbb{Q}(\sqrt[4]{2}, i)$, find a norm of $1 + \sqrt[4]{2} + i$.

Proof. Since $\mathbb{Q}$ is perfect, K is separable. And, a polynomial $x^4 - 2$ has four roots $\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}, -\sqrt[4]{2}i$.

These are all contained in K , and the splitting field of $x^4 - 2$ must contain $\sqrt[4]{2}$ and $\sqrt[4]{2}i/\sqrt[4]{2} = i$,

so K is finite normal extension of $\mathbb{Q}$, so $\text{Gal}(K/\mathbb{Q})$ is the Galois group. Since $\sigma \in \text{Gal}(K/\mathbb{Q})$ maps i to $\pm i$, $\sqrt[4]{2}$ to $\sqrt[4]{2} \cdot i^k$ ($k=0,1,2,3$), so, the norm is

$$\begin{aligned} & (1+i+\sqrt[4]{2})(1+i-\sqrt[4]{2})(1+i+\sqrt[4]{2}i)(1+i-\sqrt[4]{2}i)(1-i+\sqrt[4]{2})(1-i-\sqrt[4]{2})(1-i+\sqrt[4]{2}i)(1-i-\sqrt[4]{2}i) \\ &= ((1+i)^2 - \sqrt{2})((1+i)^2 + \sqrt{2})((1-i)^2 - \sqrt{2})((1-i)^2 + \sqrt{2}) \\ &= ((1+i)^4 - 2)((1-i)^4 - 2) \\ &= (1+i)^4 \cdot (1-i)^4 - 2((1+i)^4 + (1-i)^4) + 4 \\ &= 2^4 + 4 - 2(\underbrace{(1+i)^4}_{-4} + \underbrace{(1-i)^4}_{-4}) \\ &= 2^4 + 4 + 8 \\ &= 36. \end{aligned}$$

$\begin{array}{cc} 1+2i-1 & 1-2i-1 \\ = 2i & = -2i \\ 4i^2 = -4 & -4 \end{array}$

Let K/F be an Extension field of degree n , and let

$$\beta = \{\alpha_1, \dots, \alpha_n\}$$

be a basis for K over F .

Prop. K is isomorphic to a subfield of $M_{nn}(F)$.

Proof. Consider a map $K \rightarrow L(K, K): \alpha \mapsto T_\alpha$

where $T_\alpha(x) = \alpha \cdot x$. By simple calculation, T_α is linear, so this map is well-defined. Moreover, this map is injective linear map, because: given $\alpha, \beta \in K$ and $c \in F$,

$$T_{c\alpha + \beta}(x) = (c\alpha + \beta) \cdot x = c\alpha x + \beta x = c \cdot T_\alpha(x) + T_\beta(x).$$

And, $T_\alpha = 0$ iff $\alpha = 0$, so kernel of this map is trivial so injective.

Relates $L(K, K)$ and $M_{nn}(F)$ is omitted.

$$(1-t)^3 - 2(1-t) - 2(-1-t) + 2t$$

Prop. The characteristic polynomial of $[T_\alpha]_\beta$ has a root as α .

Proof. Let $\alpha \in K$ be fixed, with $\alpha \neq 0$. Put $A = [T_\alpha]_\beta$.

By definition, $T_\alpha(\alpha_i) = \alpha \cdot \alpha_i = \sum_{j=1}^n A_{ji} \cdot \alpha_j$.

Let $V = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in K^n$. Then, $A^t V = \begin{pmatrix} \alpha \cdot \alpha_1 \\ \vdots \\ \alpha \cdot \alpha_n \end{pmatrix} = \alpha \cdot V$

$$(1-t)^3 = 1 - t^3 - 3t(1-t) = (1-t)^3 + 3t^2 - 3t$$

or $(\alpha I - A^t)V = 0$. since $V \neq 0$, $\det(\alpha I - A^t) = 0$.

Thus proved. ~~Q~~

$$= (1-t)^3 - 2(1-t) - 2(1-t) + 4 - 2(1-t) + 6$$

Example. Find a monic minimal polynomial satisfied by $\sqrt[3]{2}$ and $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

Let $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$ be the basis for $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$.

Put $\alpha = 1 + \sqrt[3]{2} + \sqrt[3]{4}$. Then,

$$T_\alpha(1) = 1 + \sqrt[3]{2} + \sqrt[3]{4}, \quad T_\alpha(\sqrt[3]{2}) = 2 + \sqrt[3]{2} + \sqrt[3]{4}, \quad T_\alpha(\sqrt[3]{4}) = 2 + 2\sqrt[3]{2} + \sqrt[3]{4}$$

So, $[T_\alpha]_\beta = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, and $\det([T_\alpha]_\beta - tI) = -t^3 + 3t^2 + 3t + 1$